



The Alveolar Bone

Structure, Function, and Clinical Significance

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Learning Objectives

- Define the Alveolar Process and differentiate between Alveolar Bone Proper and Supporting Alveolar Bone.
- Describe the chemical composition of the bone matrix (organic vs. inorganic).
- Identify the histology and functions of osteoblasts, osteocytes, and osteoclasts.
- Explain the mechanisms of bone modeling and remodeling.
- Discuss the clinical significance of alveolar bone in tooth movement and extraction.



Alveolar bone is the mineralised connective tissue that supports and protects the teeth and is similar to bone in other sites.

The extracellular matrix is composed of hydroxyapatite crystals embedded in a collagenous matrix.

Bone can be classified according to its development (endochondral, intramembranous), histology (compact, cancellous) and adult lamellar or young woven.

Similar to bone elsewhere, it also provides for muscle attachment and contains bone marrow. In order to permit remodelling as functional demands change, bone has the significant property of 'plasticity'.

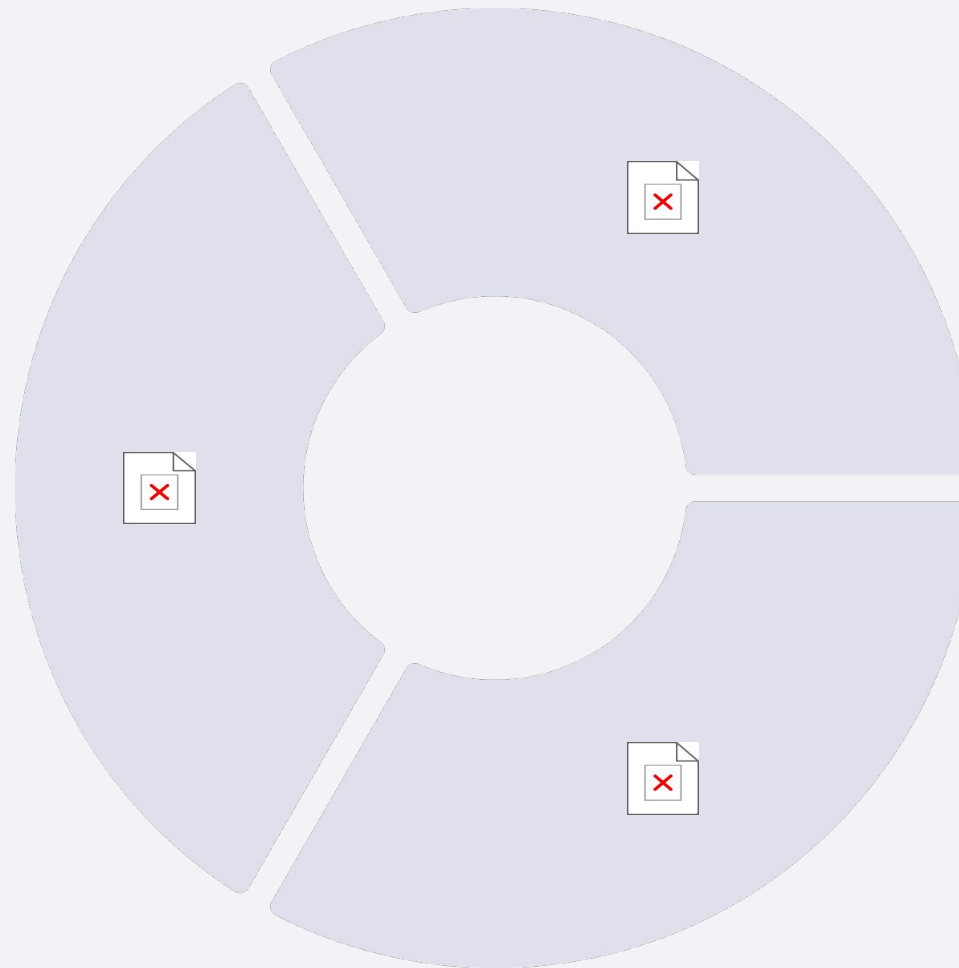
There are four main types of bone cells: three (osteoblast, osteocyte and bone-lining cell) are mesenchymal in origin while the fourth (osteoclast) is haemopoietic in origin. In addition, as bone remodels throughout life, there is a continuous need for stem/osteoprogenitor cells.

The basic structural unit of bone is the osteone (Haversian system). In addition to forming bone, the osteoblast is also important in controlling the activity of osteoclasts.

The processes of bone resorption and formation at sites of remodelling do not occur randomly. There must be tight control to ensure a balance between the two processes (coupling), as any disruption of this balance can lead to bone disease. Compact bone has a slower rate of turnover than cancellous bone. Osteocytes are thought to be the strain-detecting cells that control the remodelling system

Alveolar Bone Matrix Composition

Hydroxyapatite Crystals
Inorganic component providing compressive strength and rigidity



Type I Collagen

Constitutes 90% of organic matrix, arranged in lamellae for tensile strength

Ground Substance

Proteoglycans and glycoproteins facilitating mineralization and cell signaling

Alveolar bone is a specialized composite material where the **hydroxyapatite-collagen** matrix provides the mechanical properties necessary to resist masticatory forces while maintaining the capacity for remodeling.

BASIC FEATURES OF BONE

- Matrix of bone is a composite material of inorganic mineral and an organic matrix.
- Inorganic component is hydroxyapatite – $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$. The tiny crystals that impregnate and surround collagen fibres provide rigidity and resistance to compression.
- Organic component is ~90% type I collagen that provides strength and flexibility.
- Remaining 10% of organic component is complex mixture of proteins, including growth factors, osteocalcin, osteonectin, osteopontin, glycoproteins.
These proteins are all produced by osteoblasts, along with type I collagen.
- Unmineralised matrix of bone is termed 'osteoid'.
- Mineralisation of osteoid is dependent on the hormonally active form of vitamin D (1,25-dihydroxyvitamin D).

Cell Types in Alveolar Bone

Cell Type	Origin & Function	Key Characteristics
Osteoblasts	Mesenchymal stem cells; secrete osteoid (unmineralized bone matrix)	Regulate osteoclasts via RANKL signaling; line bone surfaces during formation
Osteocytes	Trapped osteoblasts embedded within mineralized matrix	Most numerous (>90% of bone cells); mechanosensory network via canaliculi
Osteoclasts	Monocyte/macrophage lineage; multinucleated giant cells	Responsible for bone resorption; create Howship's lacunae on bone surfaces

Box 13.2 Comparison of the main features of osteoblast, osteocytes and osteoclasts

Osteoblasts

Mesenchymal in origin

Lie on surface of forming bone

Secrete and mineralise organic matrix

Incorporated into bone matrix as osteocytes

Rich in alkaline phosphatase.

Early expression of the nuclear transcription factor core binding factor 1

Cbfa1 (also called Runx2)

Rich in endoplasmic reticulum and Golgi apparatus

Osteocytes

Mesenchymal in origin

Lie within bone and surrounded by bone matrix

Most abundant cells in bone

Mesenchymal in origin (represent entombed osteoblasts)

Communicate with each other and with osteoblasts via numerous fine canaliculi

Fewer intracellular organelles compared with osteoblasts

Detect strain in bone

May prevent hypermineralisation by releasing pyrophosphate

Endocrine/paracrine regulators of bone and mineral metabolism
(FGF23, Sclerostin, RANK ligand)

Osteoclasts

Haemopoietic in origin

Multinucleated giant cells that resorb bone

Motile

Specialised organ of resorption (ruffled border) adjacent to bone surface

Rich in acid phosphatase.

Ruffled border pumps acid (via H^+ -ATPase) to dissolve bone mineral and enzymes (mainly cathepsin K) to degrade collagenous matrix

Early expression of the nuclear transcription factor c-Fos.

Table 13.1 Comparison of woven and adult (lamellar) bone

Woven bone (immature fracture repair)	Lamellar bone (mature adult)
Osteocytes larger and irregularly spaced	Osteocytes smaller, flattened and regularly spaced
Randomly oriented collagen fibres	Collagen fibres arranged in regular plywood orientation
Enriched in acid phosphoproteins and BSP	Enriched in osteocalcin
Rapid matrix mineralisation	Delayed matrix mineralisation (a few days)
Forms rapidly	Forms slowly
Rapid turnover	Slow turnover

Bone Remodelling Cycle (Coupling Concept)

- Tight balance between **bone resorption & formation**
- Imbalance → **metabolic bone diseases**
- ⚠ Osteoclasts act **faster** than osteoblasts

4 Phases of Remodelling Cycle:

1. Resorption

1. Osteoclast recruitment, migration & activation
2. Bone breakdown begins

2. Reversal

1. Osteoclasts disappear (apoptosis/migration)
2. Surface occupied by **mononuclear cells**

3. Formation

1. Osteoblast recruitment & differentiation
2. New bone deposited in resorbed area

4. Resting

1. Mineralisation complete
2. Surface lined by **flattened cells**

Cellular & Molecular Control of Remodelling

Structural Dynamics:

- **Cutting cone:** osteoclastic resorption (~50 µm/day)
- **Closing cone:** osteoblastic bone formation
- **Reversal (cement) line:**
 - Marks transition
 - Rich in **osteopontin & acid phosphatase**

Cellular Interactions:

- Osteoblasts regulate osteoclasts via:
 - **RANKL, OPG, M-CSF**
- Osteoclast differentiation requires:
 - Close contact with osteoblasts

Key Regulators:

- Hormones & mediators:
 - Parathyroid hormone
 - Interleukins (IL-11)
 - Vitamin D₃

Environmental Factors:

- **pH & oxygen levels** influence resorption

Bidirectional Control:

- Osteoblasts → control osteoclast formation
- Osteoclasts → release signals for **osteoblast recruitment**

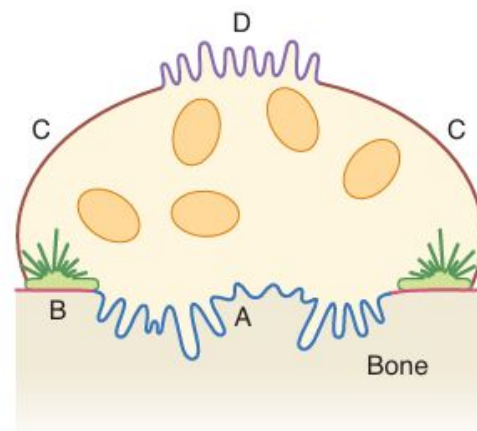


Fig. 13.21 Diagram showing four zones on osteoclast. A = ruffled border; B = sealing zone; C = basolateral zone; D = functional secretory domain.

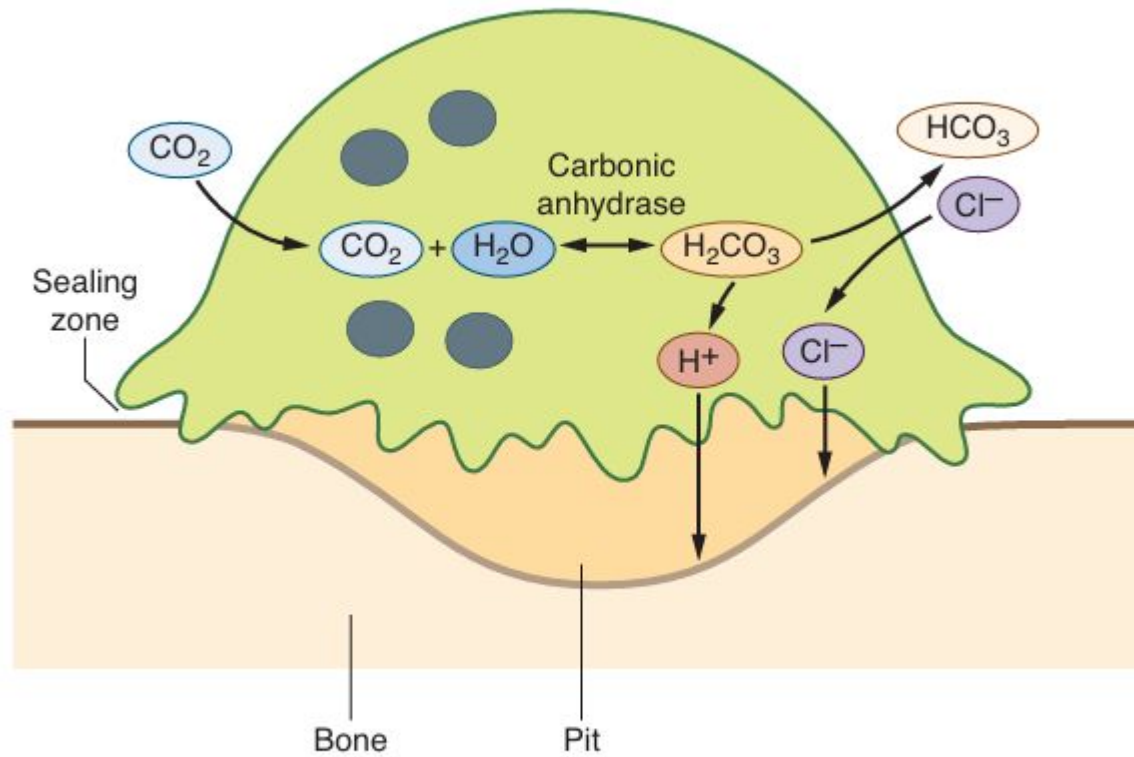


Fig. 13.25 Diagram showing action of osteoclast in resorbing bone. Courtesy Prof. T R Arnett.

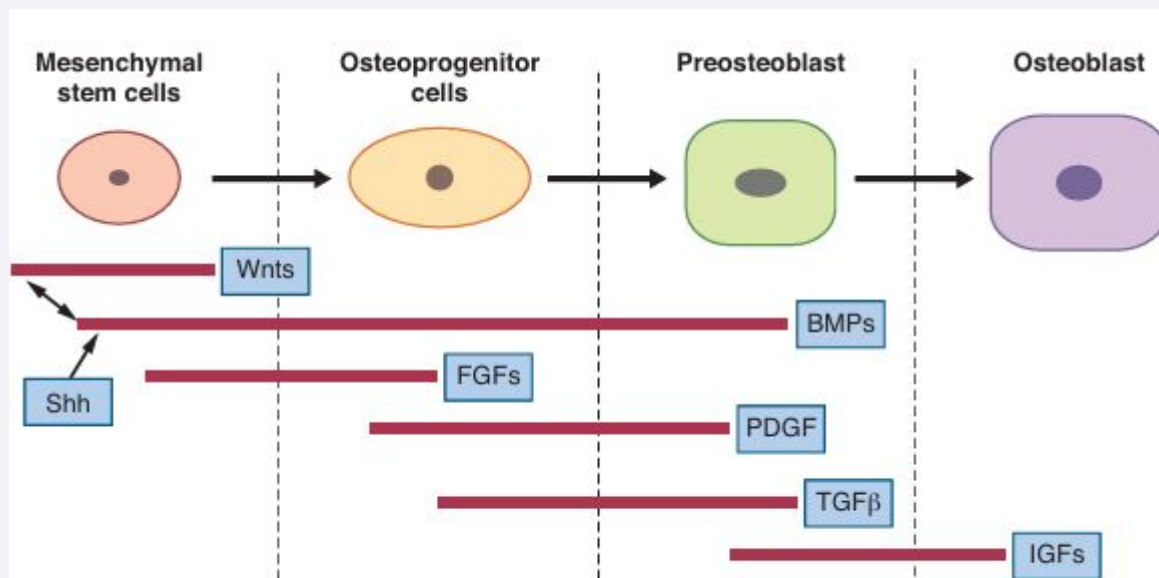
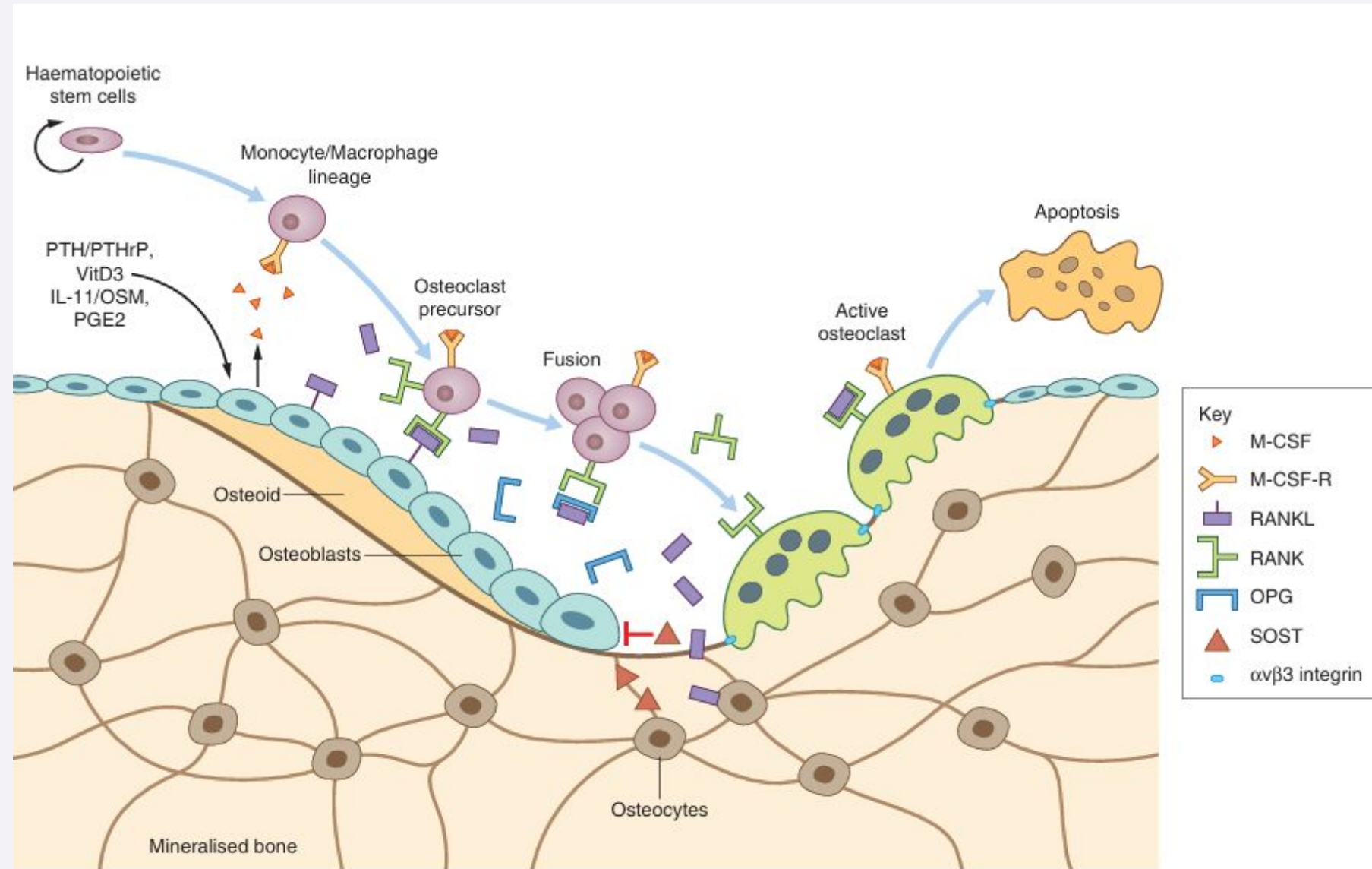


Fig. 13.26 The main stages of growth factor activity during osteoblast differentiation. BMPs = bone morphogenetic proteins; FGFs = fibroblast growth factors; IGFs = insulin-like growth factors; PDGF = platelet-derived growth factor; Shh = sonic hedgehog; TGFβ = transforming growth factor-β; Wnts = Wnt proteins. Redrawn after Hughes FJ et al 2000 *Periodontology* 2000 41: 48–72, with permission of Blackwell Publishing, a subsidiary of John Wiley & Sons Inc.



13.27 Factors controlling osteoblast–osteoclast interactions during bone remodelling. In response to calcitropic factors (e.g., PTH/PTHrP, IL-11/OSM, VitD3, PGE2), osteoblasts secrete M-CSF and RANKL which bind to M-CSF-R and RANKL, respectively, resulting in increased osteoclast differentiation, fusion, activity and survival. Osteoblasts also produce OPG which inhibits RANKL signalling. RANKL is also synthesised by osteocytes that can inhibit osteoblastic bone formation via SOST. cAMP = cyclic adenosine monophosphate; gp130 = glycoprotein 130; IL-11 = interleukin-11; M-CSF = macrophage colony-stimulating factor; M-CSF-R = M-CSF receptor; OPG = osteoprotegerin; OSM = oncostatin M; PGE2 = prostaglandin E2; PTH = parathyroid hormone; PTHrP = PTH-related protein; RANK = receptor activator of nuclear factor kappa B; RANKL = RANK ligand; SOST = sclerostin; VitD3 = 1,25-dihydroxyvitamin D3. Courtesy Dr N W A McGowan, Dr A Grigoriadis and Prof. T R Arnett.

Table 13.2 Key natural regulators of bone remodelling and mineral metabolism

Regulator	Action	Produced by
Systemic hormones		
Parathyroid hormone	↑ blood calcium ↑ bone resorption ↑ bone formation	parathyroid glands
1,25 dihydroxyvitamin D	↑ blood calcium & mineralisation ↑ bone resorption	skin (UV light)/liver/kidney
Glucocorticoids	↓ bone formation	adrenal glands
Oestrogen (& testosterone)	↓ bone resorption ↑ bone formation (smaller effect)	ovaries (& testes); adrenals
Locally-acting factors		
RANK ligand (RANKL)	↑↑ bone resorption (forms osteoclasts)	osteoblasts; osteocytes
OPG	↓ bone resorption (blocks RANKL)	osteoblasts; osteocytes
Inflammatory cytokines (interleukins 1 & 6; TNFα)	↑ bone resorption	osteoblasts; immune cells
Sclerostin	↓ bone formation (blocks Wnt signalling)	osteocytes
BMPs	↑ bone formation	osteoblasts; mesenchymal cells
VEGF	↑ bone formation (via growth of new blood vessels)	osteoblasts; many other cells
'Fundamental' (inorganic) agents		
Nitric oxide	↑ bone formation (dilates blood vessels)	blood vessel lining cells; nerves
Hypoxia (low oxygen)	↑ bone resorption (forms osteoclasts) ↓ bone formation (blocks osteoblasts)	ischaemia
Acidosis (low pH)	↑ bone resorption (activates osteoclasts)	kidney disease; tumours; diabetes; ischaemia
Pyrophosphate	↓ bone mineralisation	breakdown of ATP
Mechanical		
Strain/deformation	↑ bone formation	mechanical loading

From Amett TR 2015 *Basics of Bone Biology: Osteoporosis Review* 23: 12–16

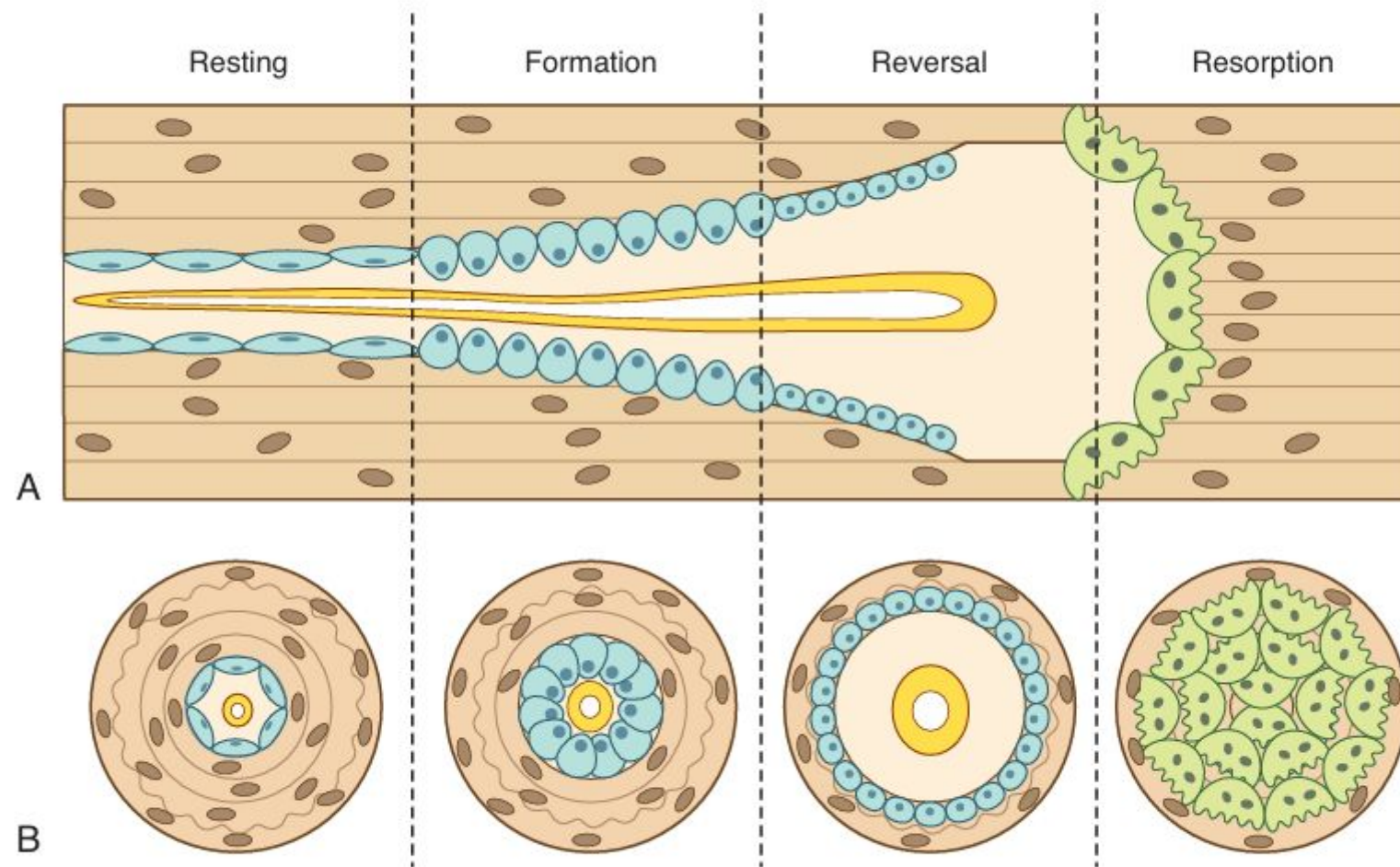


Fig. 13.31 A cutting cone. (A) Longitudinal section along a Haversian system showing the four main phases of the remodelling cycle. (B) Cross-sections of the Haversian system showing the corresponding appearance of each of the four main phases of the remodelling cycle.

Sharpey's Fibres (Alveolar Bone)

•Definition:

Extrinsic collagen fibres derived from the **periodontal ligament (PDL)**

- that insert **perpendicularly** into alveolar bone.

•Origin & Structure:

- Derived from **principal PDL fibres**
- Insert into **cribriform plate (bundle bone)**
- Thicker but **less numerous** in bone vs cementum
- Peripheral presence of **Type III collagen**

•Bundle Bone:

- Also called **cribriform plate**
- Composed of **parallel lamellae** aligned with root surface
- Named due to dense Sharpey's fibre insertion

•Mineralisation Patterns:

- **Partially mineralised:** hollow centres (seen in cervical region)
- **Fully mineralised:** project as calcified prominences into PDL

•Transalveolar Fibres:

- Pass through alveolar bone connecting:
 - Adjacent teeth (interdental bone)
 - Same tooth roots (interradicular bone)
- Absent in **cancellous bone**

•Functional Importance:

- Provide **tooth attachment & stability**
- Transmit **occlusal forces** to bone

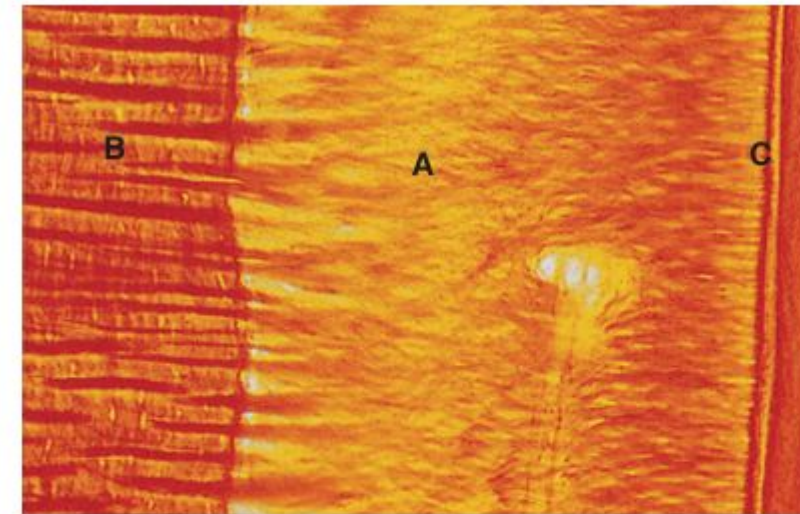


Fig. 13.41 Section of a root of a tooth showing Sharpey's fibres from the periodontal ligament (A) entering alveolar bone (B). The Sharpey's fibres in bone are seen to be thicker but less numerous than those entering the cementum on the tooth surface (C) (decalcified section; aldehyde fuchsin and van Gieson; $\times 250$). Courtesy Prof. S J Jones.

TRANSALVEOLAR FIBRES

In the cervical part of the interdental septum, where the bone type is mainly compact, Sharpey's fibres entering the bone in the mesiodistal plane may pass straight through to become continuous with similar fibres from the root of the adjacent tooth.

These are called transalveolar fibres

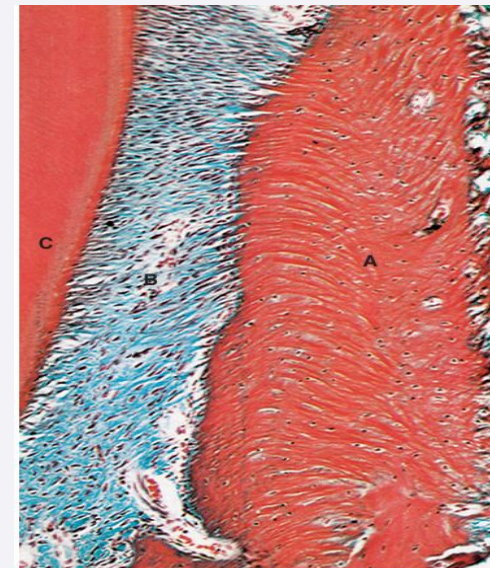
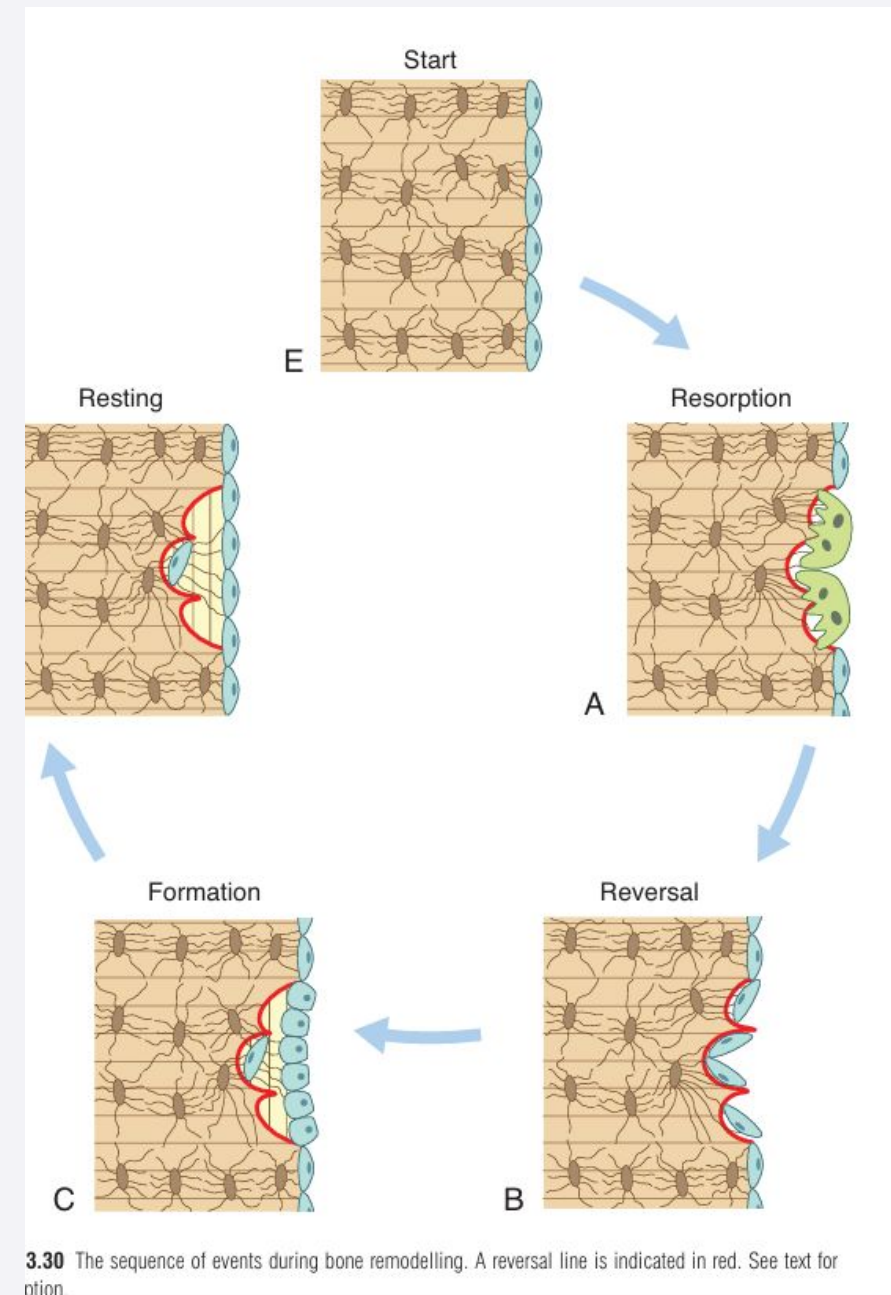


Fig. 13.45 Mesiodistal section of interdental bone (A), showing Sharpey's fibres from the periodontal ligament (B) appearing to pass more or less completely through the full thickness of bone between two adjacent teeth. C = root. (Decalcified longitudinal section; Masson's trichrome; $\times 100$.)

Bone is laid down rhythmically, which results in the formation of regular parallel lines that, because they are formed in periods of relative quiescence, are termed resting lines. Such resting lines differ biochemically from adjacent bone.

These lines are prominent in bundle bone on the distal surface of the socket wall during physiological mesial drift of the teeth (Fig. 13.47).

Bone will also contain reversal lines, representing the site of change from bone resorption to bone deposition (see Fig. 13.30). Such reversal lines will show evidence of a scalloped outline, reflecting the position of Howship's lacunae



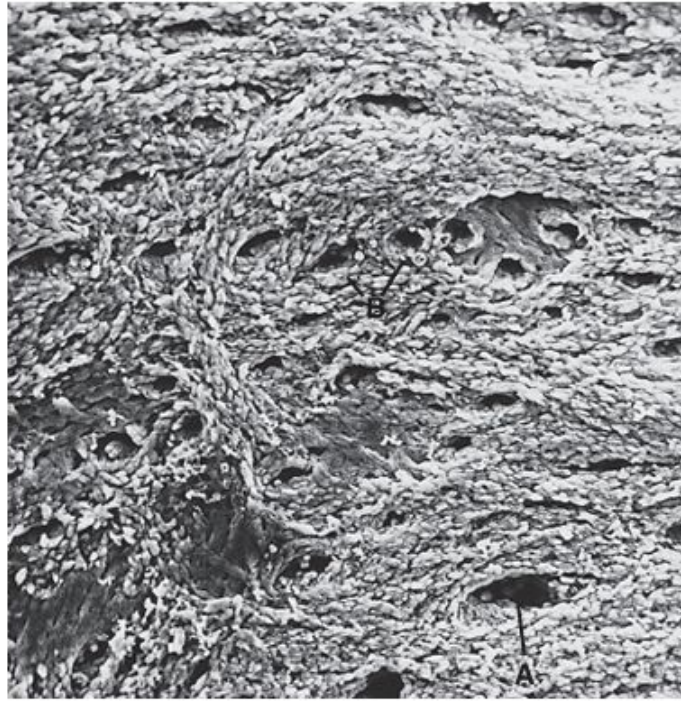


Fig. 13.35 SEM appearance of alveolar bone surface where bone formation is occurring, characterised by numerous small calcific nodules in partly mineralised intrinsic collagen, masking the underlying collagen fibrils. The larger, dark, elongated areas represent the lacunae of osteoblasts becoming entombed as osteocytes (A), while the smaller circular dark areas (B) represent the unmineralised cores of the Sharpey's fibres (anorganic preparation; $\times 550$). Courtesy Prof. S J Jones.

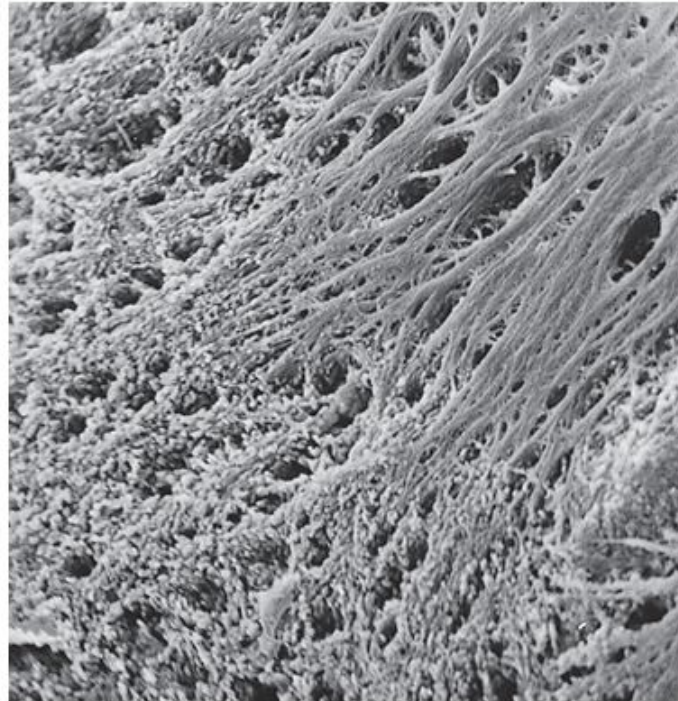


Fig. 13.36 SEM appearance of alveolar bone surface where bone formation is occurring, as evidenced by the numerous small nodules in the lower and left part of the field. However, the surface has been rendered only partly anorganic to show the collagen fibres of the extracellular matrix (partial anorganic preparation; $\times 550$). Courtesy Prof. S J Jones.



Fig. 13.37 SEM of periosteal surface of bone (lacking Sharpey's fibres) undergoing resorption. In addition to pitlike resorption lacunae (A), elongated 'snail-track' resorption lacunae are evident (B) (anorganic preparation; $\times 300$). Courtesy Prof. S J Jones.

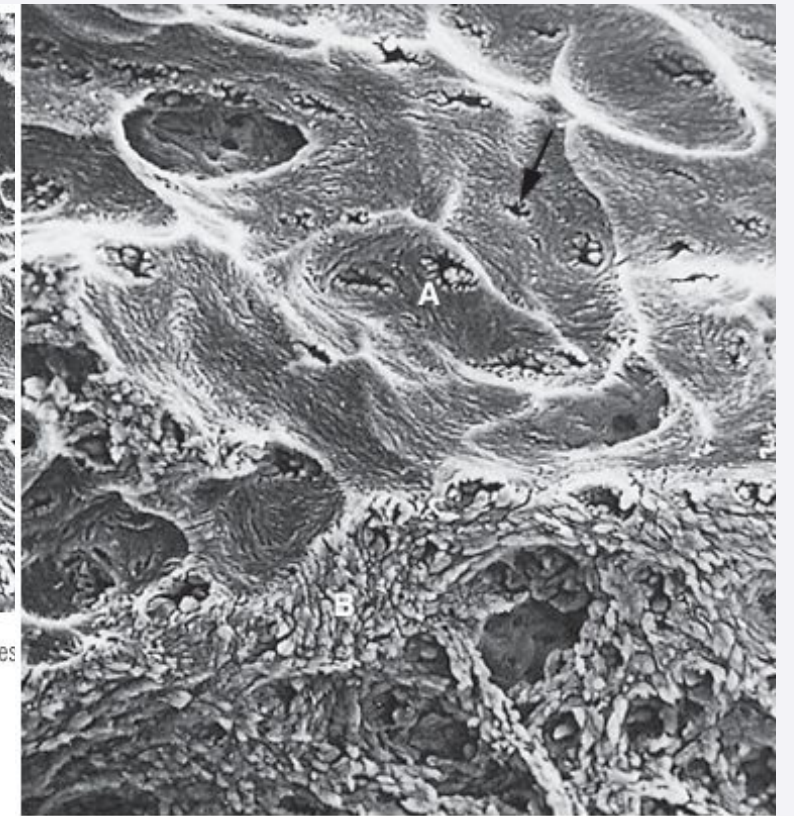


Fig. 13.38 SEM of alveolar bone surface contrasting the appearance of an adjacent area of bone formation (B). Note the presence of Sharpey's fit resorbing areas (arrowed) (anorganic preparation; $\times 300$). Courtesy Prof. S

Localised Alveolar Bone Loss (Clinical Considerations)

Key Concept: Bone depends on **functional loading**

- ↓ Occlusal load → **Bone atrophy & resorption** (e.g., after extraction)

- Dentures:

- ✓ Mild load → slows resorption
- ✗ Excess compression → **accelerates bone loss**

Orthodontic Movement:

- Bone remodeling:

- Resorption (pressure side)
- Deposition (tension side)

- Excess force → **root resorption** (osteoclasts act on dentine/cementum)

Other Causes of Bone Loss:

- **Hypoxia** (↓ blood supply): inflammation, radiation, ageing

- **Periodontal disease:**

- Plaque → inflammation → cytokines + prostaglandins → bone resorption

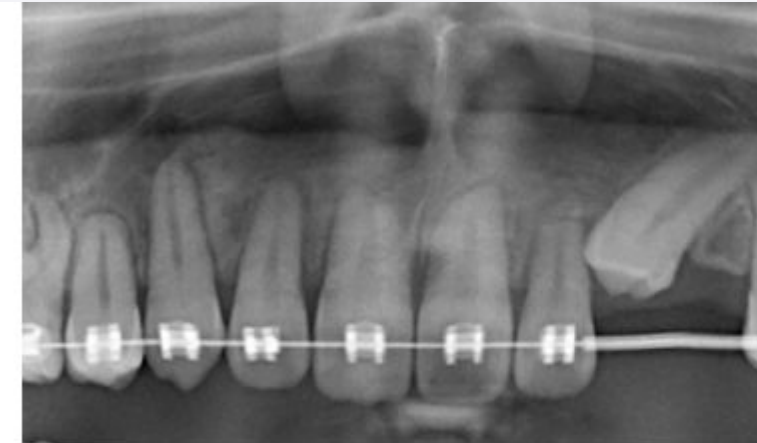


Fig. 13.49 Radiograph showing patient with fixed appliance attached to maxillary anterior teeth that has resulted in significant root resorption. Courtesy Dr J Makdissi.

Osteoporosis & Bone Metabolism

Definition:

- ↓ Bone mass, **normal composition** → ↑ fracture risk

Predisposing Factors:

- Postmenopausal **oestrogen deficiency**
- ↓ Mechanical loading
- ↑ Glucocorticoids

Sites Affected:

- Mainly **trabecular bone** (high turnover)
- Less effect on jaws (more compact bone)

Pathophysiology:

- Imbalance: ↑ **resorption** > **formation**

Therapeutic Strategies:

- ↑ OPG:RANKL ratio → ↓ osteoclast formation (e.g., oestrogen)
- Inhibit osteoclasts:
 - Calcitonin (↓ attachment)
 - Bisphosphonates (↑ apoptosis)
 - Denosumab (RANKL inhibitor)

Important Side Effect:

- Bisphosphonates → **jaw osteonecrosis**

Dental Implants & Bone Healing

Osseointegration:

- Direct **bone–implant संपर्क** (no fibrous capsule)
- Common material: **Titanium** (oxide layer critical)

Success Factors:

1. Atraumatic surgical technique
2. No micromovement
3. Implant properties (surface, shape, chemistry)

Healing Process:

- Blood clot → granulation tissue → osteoblasts
- Initial **woven bone** → **lamellar bone**
- ~17 weeks for stable interface

Limitations:

- No **periodontal ligament** → no shock absorption

Advances:

- Coating implants with:
 - Hydroxyapatite
 - Growth factors
 - Stem cells (PDL engineering – future)

Bone Grafting:

- Types:
 - Autograft (best)
 - Allograft
 - Xenograft (scaffold only)

Distraction Osteogenesis:

- Controlled bone separation → **new bone formation under tension**
- Used in craniofacial deformities & implant site preparation

